

Example 1

The time in years for a personal computer to function before breaking down is a continuous random variable with probability density function

$$f(x) = \begin{cases} \lambda e^{-x/9} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

in which λ is a constant.

- a) Determine λ .
- b) What is the expected lifetime of the computer?
- c) What is the probability that it will function less than 6 years? Discuss whether it agrees with your expectation when buying a PC.
- d) What is the name of this distribution? This distribution has a famous property. What is it? Do you think this property is realistic?

Note: Integration by parts is needed for part b). Below is a revision of integration by parts:

Integration by parts stems from the product rule for differentiation. If u and v are functions of x , then

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

Integration with respect to x gives

$$\int \frac{d}{dx}(uv) dx = \int u \frac{dv}{dx} dx + v \frac{du}{dx} dx$$

Therefore

$$uv = \int u \frac{dv}{dx} dx + \int v \frac{du}{dx} dx$$

Hence

$$\int u dv = uv - \int v du$$

Example 2

The number of attempts before achieving the first success can be considered as a discrete random variable with a probability mass function. Consider the following question.

The probability of a violin student getting the Distinction in an examination is 0.4. If the student wishes to get a Distinction, what is the expected number of attempts that the student takes the examination? Discuss the assumption(s) made behind your calculations.

Example 3

In Hong Kong, criminal cases are tried usually with a 7 person jury. In a particular case, suppose that the jurors are independent and based on the evidence, each is inclined to give a verdict of “innocent” with a probability of 0.75 in a vote, what is the probability that the vote will be a majority verdict of “innocent”?

A majority verdict is 7 – 0, 6 – 1 or 5 – 2.

Example 4

The following web site tests your typing speed: <https://www.typingtest.com/test.html>

Set up a probability model for the occurrence of mistyped words. What are your underlying assumptions?

Example 5

Suppose the average number of accidents occurring weekly on a particular stretch of a highway equals 3. Calculate the probability that there is at least one accident this week.

Example 6

Buses arrive at a specified stop at 15-minute intervals starting at 7 a.m. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7:30, find the probability that he waits

- a) less than 5 minutes for a bus;
- b) at least 12 minutes for a bus

Example 7

A tennis player has a probability of winning a tournament of 0.1. Assume that the probability of winning remain unchanged for each tournament.

- a) What is the probability that he wins 5 tournament titles after playing 20 tournaments?
- b) What is the probability that he wins the 5th tournament title at the 20th tournament he enters?
- c) How many matches is he expected to play before winning 5 tournaments?

Example 8

A buyer of components buys them in lots of size 100. It is his policy to inspect 3 components randomly from a lot and to accept the lot only if all 3 are non-defective. Suppose the manufacturing process has some systematic error such that in every lot, there is 1 defective component, what proportion of lots does the buyer reject?